

Proposal: Multi-Engine Operator Console Markov (separate)

Status: Conceptual draft for discussion — companion to [multi-engine-classroom.md](#).

Revision: 2026-09-30 — first operator Console graph (manage / alter multi-Engine setup).

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Audience: Koen / IDEA leads

Scope boundary: This graph is **only** authenticated operator manage/alter flows. Classroom learner/teacher **usage** (Kolibri / Nextcloud / Wikipedia) stays in the classroom doc and its Markov — do not mix the graphs.

Why this exists

The classroom usage Markov models people **using** apps. Operators also **manage and alter** the school's Engines: dock/eject disks, install apps, start/stop instances, redistribute load across idea-A / idea-B / idea-C, back up, erase, and manage operator accounts.

Koen's vision already says any Console can manage all Engines' apps. This document inventories **what the Console UI and Engine commands actually offer today**, then builds a separate Markov graph so later tests can walk operator flows without bloating the classroom usage graph.

Vision cite (same as classroom)

Performance is optimized by adding Appdockers and redistributing the apps over the Appdockers.

Source: [agent-engine-dev/proposals/solution-description.md](#)

Physical metaphor: **dock a disk** (USB) onto an Engine; Console **ejects** safely before unplug. Redistribution is **moveApp** / **copyApp** (and/or eject + dock elsewhere), not a separate "claim Engine" Console button.

Feature inventory (research — do not invent)

Inventoried from local clones /workspace/agent-console-dev and /workspace/agent-engine-dev (commands, components, docs). Only these land on the graph.

Console dual mode

Mode	Who	Login?	UI
User	students, teachers	No	AppBrowser — open running apps
Operator	IT / ops	Yes	NetworkTree + disk/instance management

Sources: agent-console-dev/README.md, docs/ARCHITECTURE.md, proposals/console-user-management.md.

Auth / operators (shipped)

Feature	UI	Notes
First-time setup	FirstTimeSetup	When userDB empty — create first admin
Log in	LoginForm / Account 	Username + password (bcrypt in userDB)
Log out	Account screen	Returns to user mode
Add / remove operators	OperatorManagement	Via Account → Manage Operators
Change password	Account / Settings → Account	Min 8 characters

School / multi-Engine overview (shipped)

Feature	UI	Notes
Network tree	NetworkTree	Engines + docked disks; role badges app / backup / files / empty / upgrade
Select Engine / disk	tree selection	Filters InstanceList / shows DiskView or EmptyDiskPanel
Engine discovery / connect	ConnectionManagement / Settings	Auto-discover hostnames; manual hostname; switch connected Engine
Reboot Engine	NetworkTree reboot control	Sends reboot
Operation progress / cancel	OperationProgress	cancelOperation
History / command log	HistoryPanel / CommandHistory	Observe traces

Disk roles & actions (shipped)

Disk role	Configure / act	Commands
Empty		createFilesDisk, createBackupDisk, installApp, summariseDisk/eraseDisk

Disk role	Configure / act	Commands
	Make Files Disk; Make Backup Disk; Install App; Erase	
App	Start/Stop/Open instances; Backup; Copy/Move; Add Files; Erase; Eject	startInstance, stopInstance, backupApp, copyApp, moveApp, createFilesDisk, eraseDisk, ejectDisk
Backup	Restore panel; (pure backup: no eject)	restoreApp; createBackupDisk already done
Files	Share name / availability text; eject rules	createFilesDisk (add role); mount into opted-in apps is Engine-side
Upgrade	Badge only in tree	Upgrade ops appear in operationDB when an Upgrade Disk is processed — no separate Console “Upgrade” button inventoried
System	No eject / no erase	Protected

Instance actions (shipped — InstanceRow)

- **Start / Stop / Open** ↗
- **Backup** (picker when Backup Disks exist) → backupApp
- **Copy / Move** (drag-drop desktop; MobileCopyMoveSheet on mobile) → copyApp / moveApp

Dock / eject (shipped + physical)

Action	Where	Notes
Physical dock	USB plug into Engine	Disk appears in shared store / NetworkTree — not a Console click
Eject	NetworkTree / disk → EjectConfirm	ejectDisk <diskId> — stops instances, unmounts; then operator unplugs
Undock (store)	Result of eject / physical removal	No separate “Undock” button beyond Eject

Explicitly not Console UI today (flagged)

Item	Status
Claim Engine / pool claim	Not a Console feature. Fleet claim scripts are test/Ops infra (PI_FLEET.md); classroom doc already lists them out of scope. Omitted from graph.
Dedicated uninstall / removeApp button	No Console command helper. Removing an app’s instances is via erase disk (or stop + leave Stopped). Do not invent a remove-app node.
buildEngine provisioning	Engine CLI / Ops — not daily Console Markov.

Item	Status
Automatic “redistribute wizard”	Vision text only; Console redistribution = moveApp / copyApp + physical re-dock.

Confidence: High for commands in `src/store/commands.ts` and panels under `src/components/` listed above. Medium for Upgrade Disk operator *workflow* (badge + operation kinds exist; no dedicated upgrade click script). Low/N/A for claim — confirmed absent from Console.

Markov operator graph (conceptual)

Idea

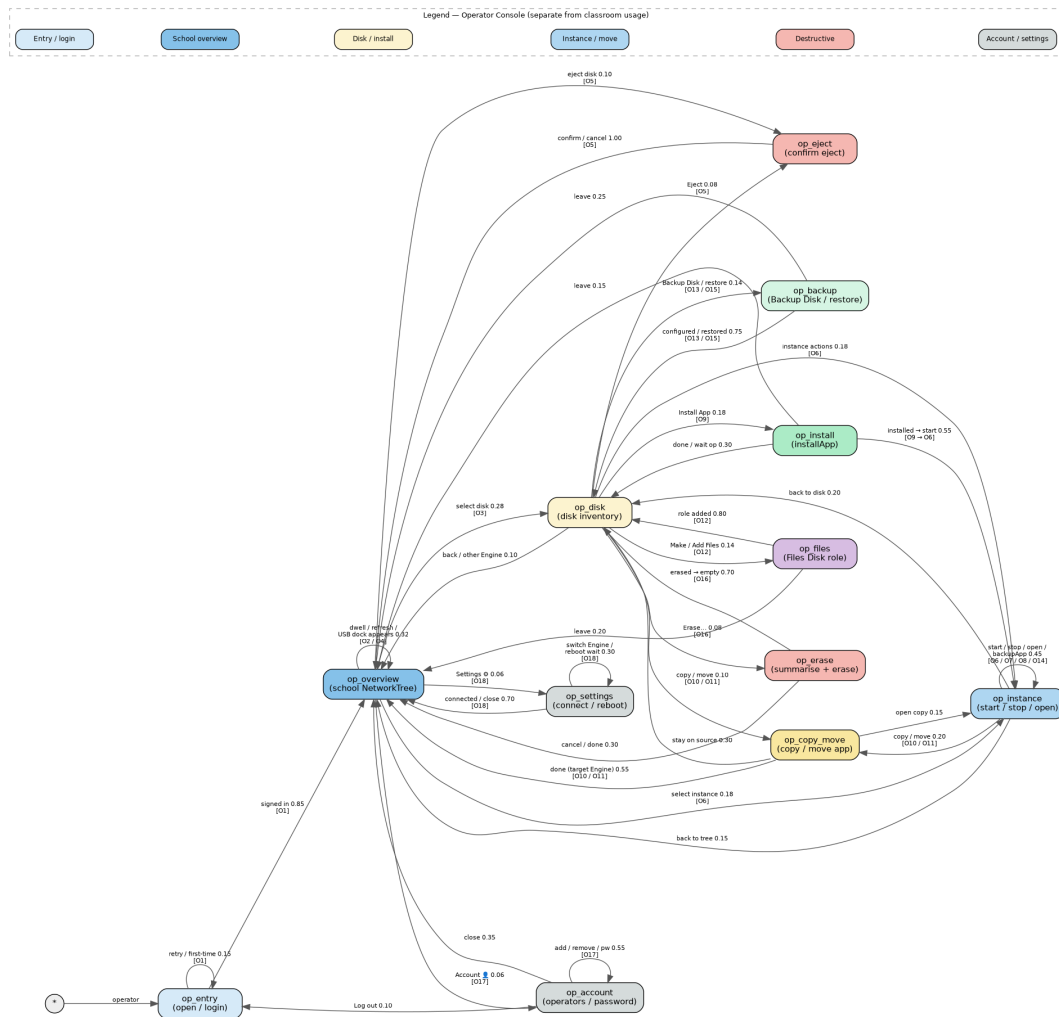
- **Nodes** = coarse operator places in the Console (not one node per click).
- **Transitions** = manage actions with placeholder probabilities + scenario IDs 01... on edges.
- Click scripts under [Detailed UI scenarios](#) expand nodes.
- **Separate files** from classroom usage: [multi-engine-operator-markov.dot](#) (+ png/svg). Classroom graph untouched except a pointer.

Nodes

Node	Meaning	Scenario IDs
op_entry	Open Console, connect/discover, Log in or first-time setup	01
op_overview	Operator NetworkTree — idea-A/B/C, disks, status (hub)	02, 04 (USB dock appears)
op_disk	Selected disk — EmptyDiskPanel or DiskView inventory	03
op_eject	Eject confirm dialog	05
op_instance	Start / Stop / Open / on-demand Backup on an instance	06, 07, 08, 014
op_install	Install App onto a disk	09
op_copy_move	Copy or Move app to another disk (possibly other Engine)	010, 011
op_files	Make / Add Files Disk role	012
op_backup	Configure Backup Disk; Restore from Backup Disk	013, 015
op_erase	Summarise + typed confirm erase	016
op_account	Operators list / add / remove / change password / logout	017
op_settings	Settings: Engine Connection, reboot, About	018

Transitions (probabilities are placeholders)

Graphviz: [multi-engine-operator-markov.dot](#). Rendered:



Same graph as Mermaid (editable)

stateDiagram-v2

[*] --> op_entry: operator

op_entry --> op_overview: signed in 0.85 [01]

op_entry --> op_entry: retry / first-time 0.15 [01]

op_overview --> op_disk: select disk 0.28 [03]

op_overview --> op_instance: select instance 0.18 [06]

op_overview --> op_eject: eject disk 0.10 [05]

op_overview --> op_account: Account 0.06 [017]

op_overview --> op_settings: Settings 0.06 [018]

op_overview --> op_overview: dwelt / refresh / USB dock 0.32 [02 / 04]

op_disk --> op_install: Install App 0.18 [09]

op_disk --> op_files: Make / Add Files 0.14 [012]

op_disk --> op_backup: Backup Disk / restore 0.14 [013 / 015]

op_disk --> op_instance: instance actions 0.18 [06]

op_disk --> op_copy_move: copy / move 0.10 [010 / 011]

op_disk --> op_erase: Erase 0.08 [016]

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op_disk --> op_eject: Eject 0.08 [05]
op_disk --> op_overview: back 0.10

op_eject --> op_overview: confirm / cancel 1.00 [05]

op_erase --> op_disk: erased → empty 0.70 [016]
op_erase --> op_overview: cancel / done 0.30

op_files --> op_disk: role added 0.80 [012]
op_files --> op_overview: leave 0.20

op_backup --> op_disk: configured / restored 0.75 [013 / 015]
op_backup --> op_overview: leave 0.25

op_install --> op_instance: installed → start 0.55 [09 → 06]
op_install --> op_disk: done / wait 0.30
op_install --> op_overview: leave 0.15

op_instance --> op_instance: start/stop/open/backup 0.45 [06/07/08/014]
op_instance --> op_copy_move: copy / move 0.20 [010 / 011]
op_instance --> op_disk: back to disk 0.20
op_instance --> op_overview: back to tree 0.15

op_copy_move --> op_overview: done 0.55 [010 / 011]
op_copy_move --> op_disk: stay on source 0.30
op_copy_move --> op_instance: open copy 0.15

op_account --> op_account: add / remove / pw 0.55 [017]
op_account --> op_overview: close 0.35
op_account --> op_entry: Log out 0.10

op_settings --> op_overview: connected / close 0.70 [018]
op_settings --> op_settings: switch Engine / reboot 0.30 [018]
```

Story walkthrough (nodes → real clicks)

How this maps to tests. Each Markov **node** is a coarse place an operator can be. Inside a node, the harness expands into an **ordered UI script** — no node-per-click. Scenario IDs in [brackets] on edges name the scripts below. Probabilities are placeholders.

UI labels match Console components (NetworkTree, EmptyDiskPanel, DiskView, InstanceRow, EjectConfirm, EraseDialog, AccountScreen, SettingsPanel). Exact Playwright selectors belong in a later harness.

Multi-Engine story: operator may open Console on **any** of idea-A / idea-B / idea-C; after login the NetworkTree shows the **school mesh** (shared store). Commands are sent to the Engine that owns the target disk (dockedTo).

op_entry — open Console / sign in

Operator opens a browser (or Chrome extension) on the school LAN, lands on user-mode AppBrowser if already connected, then elevates to operator mode.

Concrete UI (scenario 01):

1. Open Console URL for any Engine (e.g. idea-A) — or Settings → connect via discovery / hostname.
2. If userDB empty → **First-time setup**: create admin username + password → lands in operator mode.
3. Else: status bar  **Account** (or Log in) → Username / Password → **Log in**.
4. UI switches to operator layout: **NetworkTree** (left) + instances/disk detail → [op_overview](#).

op_overview — school NetworkTree

Hub state. Operator sees Engines (idea-A / B / C), disks with role badges, instance status. Dwells, refreshes mentally as CRDT updates arrive, or reacts when a USB disk is docked.

Typical actions:

1. **Dwell / refresh** (~0.32) — watch status dots; wait for a new disk after physical USB dock (02 / 04).
2. **Select a disk** → [op_disk](#) (03).
3. **Select / act on instance** → [op_instance](#) (06).
4. **Eject control on a disk** → [op_eject](#) (05).
5.  **Account** → [op_account](#) (017).
6. ⚙ **Settings** → [op_settings](#) (018).

Physical dock (04): plug SSD into idea-B (or C); disk row appears under that Engine; select it → [op_disk](#). Not a Console button.

op_disk — disk inventory

Empty disk → EmptyDiskPanel cards. App/Backup/Files disk → DiskView sections (Apps, Backups, Files).

Branch by panel:

Action	→ node	ID
Install App	op_install	09
Make this a Files Disk / Add Files	op_files	012
Make this a Backup Disk / open Backups restore	op_backup	013 / 015
Instance Start/Stop/Open/Backup	op_instance	06...
Copy / Move instance	op_copy_move	010 / 011
Erase this disk...	op_erase	016
Eject	op_eject	05

op_eject — confirm eject

Scenario 05: Eject control → EjectConfirm → confirm → ejectDisk → disk undocks in tree → operator may physically unplug → back to op_overview.

op_instance — start / stop / open / backup

Scenarios 06 / 07 / 08 / 014: On InstanceRow: **Start**, **Stop**, **Open** ↗, **Backup** (disk picker). Stay in-node for several toggles; leave to copy/move, disk, or overview.

op_install — installApp

Scenario 09: Empty (or eligible) disk → **Install App** → pick app from catalog → optional name → Install → wait OperationProgress → often **Start** (06).

op_copy_move — redistribute

Scenarios 010 (copy) / 011 (move): Drag instance to another App Disk (possibly on another Engine in the tree) or use mobile sheet **Copy** / **Move**. Vision “redistribute apps over Appdockers” = these commands (+ eject/dock). Target disk must be docked to the Engine that receives the command.

op_files — Files Disk role

Scenario 012: Make this a Files Disk (empty) or **Add Files** (app/backup disk) → share name (default School Files) → createFilesDisk. Optional erase-then-files path uses op_erase first.

op_backup — Backup Disk / restore

Scenarios 013 / 015: Empty → **Make this a Backup Disk** (mode: on-demand / immediate / scheduled + link instances). On a Backup Disk section → pick archive → target App Disk → **Restore** (restoreApp). On-demand backupApp from an instance can also start from op_instance (014).

op_erase — erase

Scenario 016: Erase this disk... → EraseDialog → Engine summariseDisk → type exact label → eraseDisk → disk becomes empty → often return to op_disk EmptyDiskPanel.

op_account — operators / password

Scenario 017:  → Manage Operators (add / remove) / change password / **Log out** → op_entry.

op_settings — connect / reboot

Scenario 018:  → Engine Connection (discover / Connect / manual hostname) / Account tab / About. Reboot from NetworkTree may be modelled as leaving overview into a wait on settings/overview self-loop after reboot.

Detailed UI scenarios (expand nodes into clicks)

These are **test scripts**, not extra Markov nodes. Preload: three Engines visible in store (idea-A/B/C), at least one empty USB disk, one App Disk with a Stopped instance, one Backup Disk when testing restore.

Scenario ID	Markov node(s)	Matches
01	* → op_entry → op_overview	Connect + login / first-time
02	op_overview	Inspect school tree (dwell)
03	op_overview → op_disk	Select disk inventory
04	op_overview (self) → op_disk	Physical USB dock appears
05	op_overview / op_disk → op_eject → op_overview	Eject confirm
06	→ op_instance	Start instance
07	op_instance	Stop instance
08	op_instance	Open app (ops check)
09	op_disk → op_install	Install App
010	→ op_copy_move	Copy app to another disk
011	→ op_copy_move	Move app (redistribute)
012	op_disk → op_files	Make / Add Files Disk
013	op_disk → op_backup	Make Backup Disk
014	op_instance	On-demand backupApp
015	op_disk → op_backup	Restore from Backup Disk
016	op_disk → op_erase	Erase disk
017	op_overview → op_account	Manage operators / password
018	op_overview → op_settings	Settings / switch Engine / reboot

Scenario 01 — Operator opens Console and signs in

Role: operator · **Engines:** any Console (idea-A / B / C).

Markov: * → op_entry → op_overview.

1. Browser → Console on school LAN.
2. If needed: Settings → discover / Connect to an Engine.
3.  → username / password → Log in (or First-time setup).
4. Assert: NetworkTree visible with multiple Engines when mesh is up.

Scenario 02 — Inspect school overview

Markov: op_overview self-loop.

1. Expand idea-A, idea-B, idea-C in NetworkTree.
2. Note disk badges (app / backup / files / empty) and instance status dots.
3. Dwell (CRDT updates) without changing selection.

Scenario 03 — Select disk inventory

Markov: op_overview → op_disk.

1. Click an empty disk under idea-B → EmptyDiskPanel cards visible.
2. Or click an App Disk → DiskView Apps / Files / etc.

Scenario 04 — Physical dock appears

Markov: op_overview (USB event) → select new disk → op_disk.

1. From op_overview, note disk list.
2. (*Hardware / fleet harness*) Plug formatted or unformatted SSD into idea-C.
3. Wait until new disk row appears under idea-C → select it → op_disk.

Scenario 05 — Eject disk

Markov: → op_eject → op_overview.

1. On a non-system, non-pure-backup disk with device: Eject.
2. Confirm in EjectConfirm.
3. Assert: disk undocked / removed from that Engine's docked list; instances stopped.

Scenario 06 — Start instance

Markov: → op_instance.

1. Select Stopped instance on idea-B App Disk.
2. **Start** → wait until Running (or Error + History).
3. Optional: **Open** ↗ (08).

Scenario 07 — Stop instance

Markov: op_instance.

1. On Running instance → **Stop** → Stopped / Docked as applicable.

Scenario 08 — Open app (ops check)

Markov: op_instance.

1. Running instance → **Open** ↗ → app URL loads (smoke only; deep app use is classroom graph).

Scenario 09 — Install App

Markov: op_disk → op_install → often op_instance.

1. Empty disk on idea-C → **Install App** → pick Kolibri (or Nextcloud) → Install.
2. Wait operation → **Start** (06).
3. Assert: instance appears in every Console's catalog (classroom S1 / reason 1).

Scenario 010 — Copy app

Markov: → op_copy_move.

1. From instance on idea-B App Disk, Copy to App Disk on idea-C (drag or mobile sheet).
2. Wait copyApp operation → new InstanceID on target.

Scenario 011 — Move app (redistribute)

Markov: → op_copy_move.

1. Move demanding Kolibri from idea-A disk to idea-B disk (same InstanceID).
2. Assert: backup links intact; source cleaned; catalog still unified.

Scenario 012 — Make / Add Files Disk

Markov: op_disk → op_files.

1. Empty disk → **Make this a Files Disk** → share name School Files (or custom) → submit.
2. Or App Disk → **Add Files** → same.
3. Assert: files badge; availability text updates when opted-in apps run.

Scenario 013 — Make Backup Disk

Markov: op_disk → op_backup.

1. Empty disk → **Make this a Backup Disk** → mode on-demand → select instance(s) → Configure.
2. Assert: backup badge.

Scenario 014 — On-demand backup

Markov: op_instance.

1. Running (or eligible) instance → **Backup** → pick Backup Disk → backupApp.
2. Watch OperationProgress; cancel only if testing cancel path.

Scenario 015 — Restore from Backup Disk

Markov: op_disk → op_backup.

1. Select Backup Disk → Restore panel → pick instance archive → target App Disk → confirm Restore.

Scenario 016 — Erase disk

Markov: op_disk → op_erase → op_disk (empty).

1. **Erase this disk...** → wait summary → type exact label → confirm.
2. Assert: disk empty; prior instances gone from store for that disk.

Scenario 017 — Manage operators

Markov: op_overview → op_account.

1.  → Manage Operators → Add operator / Remove / Change password.
2. Optional: Log out → op_entry.

Scenario 018 — Settings / switch Engine / reboot

Markov: op_overview → op_settings.

1.  → Engine Connection → Connect to another discovered Engine (or manual hostname).
2. Optional: NetworkTree **Reboot** on a selected Engine → wait reconnect → op_overview.

Composite walks (multi-Engine alter)

Walk	Story	Scenario chain
Add capacity	New empty disk on idea-C → install Kolibri → start	04 → 09 → 06
Redistribute load	Move heavy instance idea-B → idea-C	03 → 011
Safe disk carry	Eject on idea-B → (unplug) → dock on idea-A	05 → 04 → 03
Files for Nextcloud	Make Files Disk on idea-A → start Nextcloud	012 → 06
Backup drill	Make Backup Disk → backupApp → restore elsewhere	013 → 014 → 015

Relation to classroom doc

Document	Graph	Audience
multi-engine-classroom.md	multi-engine-markov.*	Learners + teachers using Kolibri / Nextcloud / Wikipedia
This file		

Document	Graph	Audience
	multi-engine-operator-markov.*	Operators managing Engines / disks / instances

Classroom scenarios S1 / S7 mention operator dock/start only as story context; their scripts stay usage-side. Operator click paths live here (0*).

Sources consulted

Source	Use
agent-console-dev/src/store/commands.ts	Full Console→Engine command surface
agent-console-dev/src/components/*	NetworkTree, EmptyDiskPanel, DiskView, InstanceRow, EjectConfirm, EraseDialog, AccountScreen, OperatorManagement, SettingsPanel, RestorePanel, MobileCopyMoveSheet
agent-console-dev/docs/ARCHITECTURE.md	Dual-mode UI, auth, discovery
agent-console-dev/src/store/diskRoles.ts	Role badges, eject rules, Files availability
agent-engine-dev/docs/COMMANDS.md	Engine command semantics (install/start/stop/eject/erase/files/backup/copy/move/reboot)
agent-engine-dev/proposals/solution-description.md	Vision: autofind + redistribute
agent-engine-dev/proposals/install-app.md, copy-move-app.md, backup-disk.md	Merged proposal behaviour
PI_FLEET.md	Confirms fleet claim is test infra , not Console UI
multi-engine-classroom.md	Conventions for nodes / scenario IDs / story walkthrough

Out of scope / speculative

- Playwright harness wiring (later; same duration-tests family as classroom).
- **Claim Engine** Console UX — **does not exist**; do not add without a real design.
- Automatic redistribute wizard beyond copy/move + physical dock.
- Upgrade Disk one-click Console flow (badge + operation kinds only — **speculative** as a dedicated scenario until UI confirms).
- buildEngine provisioning walks.
- Messaging Koen — parent/Steve owns review handoff.

Ask of Koen

1. Confirm operator graph stays **separate** from classroom usage Markov.
2. Confirm node coarseness (12 nodes) and scenario IDs 01–018.
3. Confirm omission of **claim Engine** from the graph (not Console UI).
4. Prioritise which composite walks (add capacity / redistribute / eject-carry / backup drill) to deepen first.